Chathumi Navodya Sathsarani

BSc in Applied Data Science Communication (Undergraduate)

+94 70 339 4005 wpcnsperera@gmail.com

No.36/A, Horahena, Hokandara East, Sri Lanka

LinkedIn GitHub Portfolio

Professional Summary

Applied Data Science undergraduate with hands-on experience in data analytics, machine learning, and building data-driven systems. Skilled in Python, SQL, Power BI, and data modeling, with experience in ETL workflows, interactive dashboards, and predictive solutions. Interested in opportunities across Data Analysis, Data Science, Data Engineering, and AI/ML-focused roles.

Experience

Sampath Bank PLC
Intern – Recoveries Department Jun 2023 – Feb 2024
Analyzed repayment and overdue account data using Excel to support loan recovery tracking and reporting. Assisted with client follow-up workflows and maintained structured financial records, gaining exposure to credit risk analysis and data-driven operational reporting.

Education

Bachelor of Science in Applied Data Science & Communication 2024 – Present
General Sir John Kotelawala Defence University (KDU), Ratmalana

Certificate Course in Computer Applications May 2023 – Nov 2023
University of Colombo, Sri Lanka

Certificate Course in English Language (Intermediate) Apr 2023 – Oct 2023
University of Colombo, Sri Lanka

Skills

- **Programming:** Python, SQL, R
- **Machine Learning & AI:** Scikit-learn, TensorFlow, NLP, RAG Systems, Model Evaluation, Feature Engineering
- **Data Engineering & Analytics:** Data Preprocessing, ETL Pipelines, Data Modeling (Star Schema), EDA
- **Visualization & Dashboards:** Power BI, DAX, Plotly, Dash, Streamlit
- **Tools & Technologies:** Git, REST APIs (Gemini API), OCR, SQLite , Microsoft Power Platform (Basic)
- **Cloud:** Microsoft Azure (Basic)
- **Microsoft Tools:** Excel, PowerPoint, Word
- **Soft Skills:** Communication, Teamwork, Problem Solving

Projects

- **Policy AI Assistant – Generative AI Policy Analysis System** [GitHub] [Demo] Mar 2026
  Python, Streamlit, NLP, TF-IDF, Gemini API, HTML, CSS, JavaScript
  Developed a RAG-based AI system to automate policy document analysis, enabling structured summarization, scenario-based draft generation, and context-aware Q&A with retrieval grounding.
- **QuizCraft AI – AI Quiz Generation Platform** [GitHub] [Demo] Jan 2026
  Python, Streamlit, SQLite, Gemini API, NLP, Data Visualization
  Built an AI-powered quiz generation platform that converts documents into MCQs, with secure authentication, performance tracking, and real-time analytics dashboards.

- **Flood Impact Prediction & Relief System** [GitHub] Dec 2025
  Python, Scikit-learn, Streamlit, Pandas, Matplotlib, Plotly
  Developed an end-to-end machine learning system to predict flood occurrence, estimate inundation impact, and prioritize relief response through an interactive dashboard.
- **UK Visa Sponsorship Analysis** [GitHub] Nov 2025
  SQL, Power BI, DAX, Data Modeling
  Designed a star schema data model and developed interactive Power BI dashboards using advanced DAX measures to analyze UK migration trends from cleaned datasets.
- **Fake Job Detection System** [GitHub] Oct 2025
  Python, NLP, TF-IDF, OCR, SVM, SHAP, Streamlit
  Built an AI-powered fraud detection system that analyzes job postings from text and images using OCR and TF-IDF, providing explainable predictions with risk scoring.
- **Tourism Data Clustering System** [GitHub] May 2025
  R, K-means Clustering, ggplot2, dplyr, factoextra, Data Preprocessing
  Performed clustering analysis on tourism data to segment user behavior and generate insights for decision-making.
- **Student Performance Prediction Dashboard** [GitHub] Apr 2025
  Python, Pandas, Scikit-learn, Mlxtend, Plotly, Dash
  Built a data pipeline combining association rule mining and logistic regression to analyze and predict student performance, with an interactive dashboard for real-time insights.

Certifications

- **AI/ML Engineer – Stage 2** — SLIIT Jan 2026
- **AI/ML Engineer – Stage 1** — SLIIT Aug 2025
- **Data Analysis with Python** — IBM (Coursera) May 2025
- **GenAI 101 with Pieces** Feb 2025
- **Postman API Fundamentals Student Expert** Jan 2025

Leadership

AI and Data Science Club, KDU 2025 – 2026
Treasurer & Project Chairperson

- Managed financial records and budgeting for club events and initiatives
- Led planning and coordination of the **Tech Spark** event
- Collaborated with team members to organize and execute technical sessions

References

Ms. Chamodhi Wickramasinghe
Lecturer, General Sir John Kotelawala Defence University
+94 71 361 2792
wickramasinghe.bct@kdu.ac.lk

Dr. Charith Silva
Senior Lecturer, General Sir John Kotelawala Defence University
+44 79 20 206 338
charithsilva@kdu.ac.lk